STUDENT PERFORMANCE PREDICTION USING MACHINE LEARNING

A Mini Project Report

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Course: Fundamentals of AI and ML
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1. INTRODUCTION

In this project, I tried to understand how different daily habits affect a student's academic performance. Things like how many hours a student studies, how much they sleep, and their attendance can make a big difference in their marks.

So, I used a simple machine learning model to analyze these factors and predict marks. This helped me understand how data can be used to make useful predictions.

2. PROBLEM STATEMENT

The Aim of this project is to develop a machine learning model that predicts student marks based on:

- Study Hours
- Sleep Hours
- Attendance

3. OBJECTIVES

- To analyze student performance factors
- To build a simple prediction model
- To visualize the results using a graph

4. METHODOLOGY

The project follows these steps:

1. Data Collection (Manual dataset)

Dataset Table:

Study Hours	Sleep Hours	Attendance	Marks
2	6	60	50
4	7	70	60
6	8	80	70
8	7	90	90

2. Data Preprocessing
3. Model Selection (Linear Regression)
4. Model Training
5. Flowchart

6. Prediction
7. Visualization

5. IMPLEMENTATION

I used Python to build this project. Libraries like Pandas helped in handling data, Scikit-learn was used for the machine learning model, and Matplotlib was used to draw graphs.

The program takes input from the user and then predicts the marks based on the trained model.

6. RESULTS AND ANALYSIS

The model successfully predicts marks based on input values.

Observations:

- Increasing study hours improves marks
- Higher attendance leads to better performance
- Balanced sleep contributes to consistency

The model predicts marks based on input values. For example, when study hours are increased, the predicted marks also increase, showing a positive relationship.

📌 **Output:**

- Code Screenshot
- Output Screenshot
- Graph Screenshot

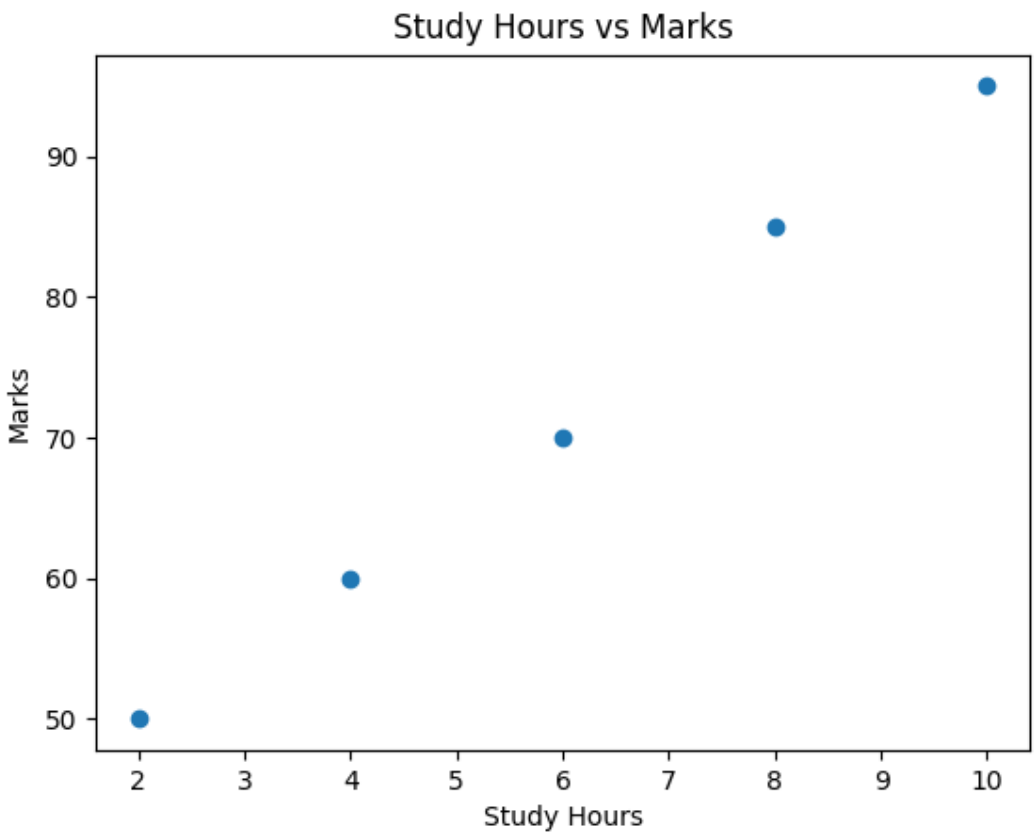
<u>Code Screenshot:</u>

```python
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

# Dataset
data = {
    'study_hours': [2, 4, 6, 8, 10],
    'sleep_hours': [5, 6, 7, 8, 9],
    'attendance': [60, 70, 80, 90, 100],
    'marks': [50, 60, 70, 85, 95]
}

df = pd.DataFrame(data)

# Features & Target
X = df[['study_hours', 'sleep_hours', 'attendance']]
y = df['marks']

# Model
model = LinearRegression()
model.fit(X, y)

# User Input
study = float(input("Enter study hours: "))
sleep = float(input("Enter sleep hours: "))
att = float(input("Enter attendance (%): "))

# Prediction
pred = model.predict([[study, sleep, att]])
print("📊 Predicted Marks:", round(pred[0], 2))

# Graph (Study Hours vs Marks)
plt.scatter(df['study_hours'], df['marks'])
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("Study Hours vs Marks")
plt.show()
```

Output Screenshot:

```
OUTPUT   DEBUG CONSOLE   PROBLEMS   TERMINAL   PORTS                                   Python

PS C:\Student-Performance-Predictor> & C:\Users\srika\AppData\Local\Microsoft\WindowsApps\python3.13.exe c:/Student-Performance-Predictor/pr
oject.py
Enter study hours: 9
Enter sleep hours: 8
Enter attendance (%): 95
C:\Users\srika\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\site-packages\s
klearn\utils\validation.py:2691: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
  warnings.warn(
📊 Predicted Marks: 89.2
```

Graph Screenshot:



7. CHALLENGES FACED

While doing this project, I faced some difficulties like installing libraries and understanding how the model works. Also, since the dataset is small, the prediction may not be very accurate.

8. FUTURE SCOPE

In the future, this project can be improved by using a larger dataset and more advanced machine learning algorithms. It can also be turned into a web application.

9. CONCLUSION

This project shows how machine learning can be used to predict student performance. It also helped me understand how study habits can affect marks and overall results. While doing this project, I got to know how data can be used to make predictions, and it gave me useful insights for improving academic performance. I also faced some small issues while working on the model and graphs, but I was able to fix them.

This project helped me learn basic machine learning concepts.

10. REFERENCES

- Python Documentation (https://docs.python.org/)
- Scikit-learn Documentation (https://scikit-learn.org/)
- Course Materials and Lecture Notes
- Online tutorials and learning resources